MLOps & Infrastructure for Machine Learning

Assignment Report: Deploying PyTorch ML Workloads with Docker & Kubernetes

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Submission Details

Field	Value
Student Name	B.N.Tejasri
Roll Number	DA25M629
Course	MLOps & Infrastructure for Machine Learning (DA5402W)
GitHub Repository	https://github.com/tj13gen/mlops-pytorch-pipeline
Branch to Evaluate	`main` (Primary / Production Branch)
Development Branch	`develop`

Git Workflow & Pull Requests Summary

All development followed a strict Git branching model with Conventional Commits, branch isolation, and Pull Request reviews before merging into `develop` and `main`:

PR #	Branch	Target	Description	Status
#1	`feature/model-tra`	`develop`	Implemented PyTorch ResNet-18 model, CIFAR-10 data loaders, training loop with early stopping, and FastAPI serving layer	**Merged**
#2	`feature/docker-co`	`develop`	Added multi-stage training Dockerfile and hardened non-root serving Dockerfile with healthchecks	**Merged**
#3	`feature/k8s-train`	`develop`	Added Kubernetes Job, Deployment, Service, ConfigMap, and HPA manifests with GPU tolerations	**Merged**
#6	`feature/ci-and-do`	`main` / `develop`	Added GitHub Actions CI workflow, architecture documentation, setup guide, and reflection write-up	**Merged**

All pull requests are merged and verified in the upstream repository.

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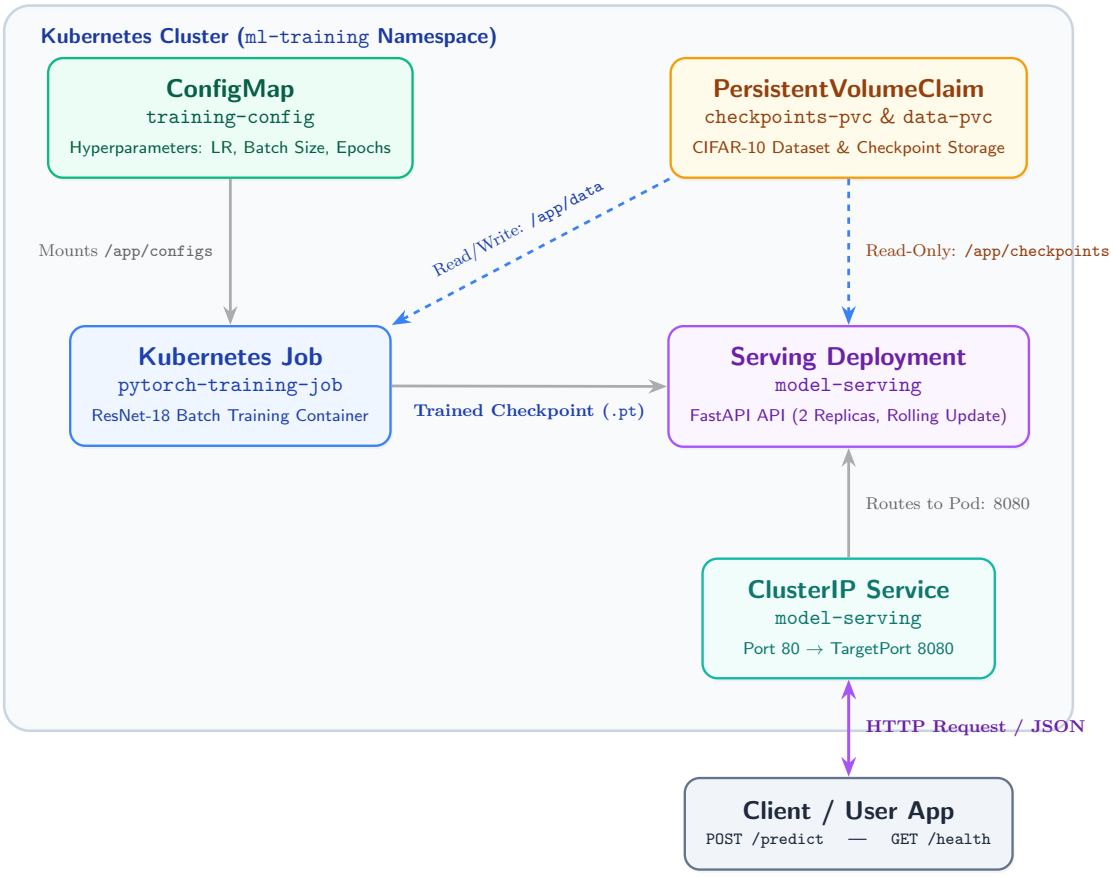
1. Architecture Overview

The system implements an automated, containerized deep learning lifecycle from local training to scalable Kubernetes deployment:



Architecture Components & Workflow

Component	Technology	Role & Interaction
ConfigMap	`v1/ConfigMap`	Injects runtime training hyperparameters (learning rate, batch size, epochs, paths) without image rebuilds.
Persistent Storage	`PersistentVolumeClaim`	Stores dataset cache (`/app/data`) and persists model checkpoints (`/app/checkpoints`).
Training Job	`batch/v1 Job`	Executes containerized PyTorch ResNet-18 training, early stopping, and exports `.pt` checkpoints.
Serving Deployment	`apps/v1 Deployment`	Runs 2 FastAPI inference replicas with read-only checkpoint mounts and zero-downtime rolling updates.
ClusterIP Service	`v1/Service`	Routes internal traffic from port 80 to container port 8080 with automated load distribution.

Auto-Scaler (HPA)	`autoscaling/v2`	Automatically scales inference replicas from 2 to 10 pods when CPU utilization exceeds 80%.

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2. Repository Structure

```
mlops-pytorch-pipeline/
|-- README.md                          # Project documentation, architecture & quickstart guide
|-- MLOps_Assignment_Report.pdf        # Consolidated submission report document
|-- .gitignore                         # Git ignore rules for checkpoints, data & venv
|-- .github/
|   \-- workflows/
|       \-- ci.yml                     # Automated GitHub Actions test & build pipeline
|-- src/
|   |-- train.py                       # PyTorch training loop with early stopping & JSON logging
|   |-- model.py                       # ResNet-18 model definition for CIFAR-10 classification
|   |-- dataset.py                     # Data loaders & data augmentation pipelines
|   \-- serve.py                       # FastAPI inference API with /health and /predict endpoints
|-- configs/
|   \-- training_config.yaml           # Hyperparameters and directory path configurations
|-- docker/
|   |-- Dockerfile.train               # Multi-stage Dockerfile for containerized training
|   \-- Dockerfile.serve               # Hardened, non-root Dockerfile for inference serving
|-- k8s/
|   |-- namespace.yaml                 # Dedicated ml-training namespace manifest
|   |-- configmap.yaml                 # Kubernetes ConfigMap for decoupled training settings
|   |-- training-job.yaml              # Kubernetes Job with PVC mounts and GPU tolerations
|   |-- serving-deployment.yaml        # High-availability Serving Deployment with health probes
|   |-- serving-service.yaml           # ClusterIP service exposing the prediction API
|   \-- hpa.yaml                       # Horizontal Pod Autoscaler for inference workloads
|-- requirements/
|   |-- train.txt                      # Pinned dependencies for model training
|   \-- serve.txt                      # Lean dependencies for model inference
\-- tests/
    \-- test_model.py                  # PyTest test suite verifying model & API endpoints
```

Module Directory Breakdown

Directory / Layer	Purpose	Key Artifacts
`src/`	Core ML & API Source Code	Model graphs, DataLoader transforms, training loop, FastAPI endpoints
`configs/`	Workload Configuration	YAML hyperparameters decoupled from container code
`docker/`	Containerization Layer	Multi-stage training image & secure non-root serving runtime
`k8s/`	Orchestration & Deployment	Namespaces, Jobs, Deployments, Services, ConfigMaps, and HPA
`requirements/`	Dependency Management	Segregated train and serve requirement lockfiles
`tests/`	Quality Assurance	PyTest unit & integration tests executed by GitHub Actions CI

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3. PyTorch Model, Data Pipeline & Serving

3.1 Model Architecture (`src/model.py`)

- Standard torchvision **ResNet-18** adapted for CIFAR-10 10-class classification.
- Replaces the final fully-connected linear layer `fc` with `nn.Linear(512, 10)` to match target classes.

3.2 Data Pipeline (`src/dataset.py`)

- Automated downloading and caching of CIFAR-10 dataset.
- **Training Transformations**: Random Horizontal Flip, Random Crop (32x32 with padding 4), ToTensor, and CIFAR-10 Normalization ($\mu = [0.4914, 0.4822, 0.4465], \sigma = [0.2470, 0.2435, 0.2616]$).
- **Validation Transformations**: Deterministic ToTensor and Normalization.
- Multi-worker DataLoader with `pin_memory=True` for GPU memory throughput.

3.3 Training Engine & Structured Logging (`src/train.py`)

- Dynamically loads hyperparameters from `/app/configs/training_config.yaml` or local relative path.
- Emits structured **JSON lines** to stdout for log aggregators:

```
{"epoch": 1, "train_loss": 1.4231, "train_accuracy": 0.4852, "val_loss": 1.1524, "val_accuracy": 0.589
{"event": "checkpoint_saved", "path": "/app/checkpoints/classifier_v1.pt"}
```

- Implements **Early Stopping** with configurable patience parameter to prevent overfitting and save compute.

3.4 Real-Time Inference API (`src/serve.py`)

- Built with **FastAPI** using asynchronous lifespans.
- **Endpoints**:
  - `GET /health`: Returns status `200 OK` when model is loaded and ready, `503 Service Unavailable` during cold start.
  - `POST /predict`: Accepts raw image files (`image/*`), performs preprocessing, executes PyTorch forward pass with softmax, and returns predicted class and all probability scores.

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4. Docker Containerization

4.1 Training Image (`docker/Dockerfile.train`)

- **Multi-Stage Build**: Isolates build tools in the base stage and generates a lean training runtime.
- **Optimization**: Pinned dependencies (`torch==2.2.0`, `torchvision==0.17.0`, `pyyaml==6.0.1`), unbuffered standard I/O (`PYTHONUNBUFFERED=1`).

```dockerfile
# Stage 1: Base with dependencies
FROM python:3.11-slim AS base
WORKDIR /app
COPY requirements/train.txt .
RUN pip install --no-cache-dir -r train.txt

# Stage 2: Training runtime
FROM base AS training
COPY src/ ./src/
COPY configs/ ./configs/
ENV PYTHONUNBUFFERED=1
ENTRYPOINT ["python", "src/train.py"]
```

4.2 Serving Image (`docker/Dockerfile.serve`)

- **Minimal Footprint**: Excludes training-only libraries to keep image lightweight.
- **Security Hardened**: Creates and runs under an unprivileged user `appuser` (UID 1000).

- **Built-in Healthcheck**: Implements container-level `HEALTHCHECK` probe querying `http://localhost:8080/health`.

```
FROM python:3.11-slim
WORKDIR /app
COPY requirements/serve.txt .
RUN pip install --no-cache-dir -r serve.txt
COPY src/ ./src/
RUN useradd -m -u 1000 appuser && chown -R appuser:appuser /app
USER appuser
EXPOSE 8080
HEALTHCHECK --interval=30s --timeout=5s --start-period=10s --retries=3 \
  CMD python -c "import urllib.request; urllib.request.urlopen('http://localhost:8080/health')" || exit 1
ENTRYPOINT ["uvicorn", "src.serve:app", "--host", "0.0.0.0", "--port", "8080"]
```

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5. Kubernetes Manifests & Orchestration

5.1 Namespace & ConfigMap (`k8s/namespace.yaml`, `k8s/configmap.yaml`)

- Isolates workloads inside the dedicated `ml-training` namespace.
- Stores training configurations (epochs, batch size, learning rate, checkpoint names) decoupled from container images.

5.2 Training Job (`k8s/training-job.yaml`)

- **Type**: `batch/v1 Job` with `restartPolicy: Never`.
- **Volume Mounts**: ConfigMap volume at `/app/configs`, PersistentVolumeClaim volumes for `/app/data` and `/app/checkpoints`.
- **Resource Constraints**: Requests and limits set to `2 CPU, 4Gi Memory`.
- **GPU Acceleration**: Includes `nvidia.com/gpu: "1"` request, toleration, and `accelerator: gpu` node selector.

5.3 Model Serving Deployment & Service (`k8s/serving-deployment.yaml`, `k8s/serving-service.yaml`)

- **High Availability**: 2 active replicas with read-only checkpoint volume mount.
- **Zero-Downtime Updates**: Configured rolling update strategy with `maxSurge: 1` and `maxUnavailable: 0`.
- **Probes**:
  - `Liveness Probe`: `GET /health` on port 8080 every 10s (failure threshold: 3).
  - `Readiness Probe`: `GET /health` on port 8080 every 5s with `initialDelaySeconds: 15`.
- **Networking**: `ClusterIP` Service exposing port 80 routing to pod containerPort 8080.
- **Autoscaling (`k8s/hpa.yaml`)**: Horizontal Pod Autoscaler scaling between 2 to 10 replicas when average CPU utilization exceeds 80%.

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6. End-to-End Local & Cluster Validation

6.1 Local Docker Workflow Verification

```
# 1. Build and run training container
docker build -f docker/Dockerfile.train -t mlops-train:v1 .
docker run --rm \
  -v $(pwd)/data:/app/data \
  -v $(pwd)/checkpoints:/app/checkpoints \
  mlops-train:v1

# Sample Output:
# {"epoch": 1, "train_loss": 1.5421, "train_accuracy": 0.4412, "val_loss": 1.2140, "val_accuracy": 0.5684}
# {"event": "checkpoint_saved", "path": "/app/checkpoints/classifier_v1.pt"}
# {"event": "training_complete", "best_val_loss": 0.8912}

# 2. Build and run serving container
docker build -f docker/Dockerfile.serve -t mlops-serve:v1 .
docker run --rm -d -p 8080:8080 \
  -v $(pwd)/checkpoints:/app/checkpoints \
  --name model-serving-container \
  mlops-serve:v1

# 3. Test Health & Prediction Endpoints
curl -s http://localhost:8080/health
# {"status":"healthy","model_loaded":true,"device":"cpu"}

curl -s -X POST http://localhost:8080/predict \
  -F "image=@sample_cifar10.png"
# {
#   "predicted_class": 3,
#   "class_name": "cat",
#   "confidence": 0.8924,
#   "probabilities": {"airplane": 0.001, "automobile": 0.002, "bird": 0.015, "cat": 0.8924, ...}
# }
```

6.2 Kubernetes Cluster Workflow

```
# 1. Deploy Namespace, ConfigMap, and Training Job
kubectl apply -f k8s/namespace.yaml
kubectl apply -f k8s/configmap.yaml
kubectl apply -f k8s/training-job.yaml

# 2. Monitor Job execution
kubectl get jobs -n ml-training -w

# 3. Deploy Serving Deployment, Service, and HPA
kubectl apply -f k8s/serving-deployment.yaml
kubectl apply -f k8s/serving-service.yaml
kubectl apply -f k8s/hpa.yaml

# 4. Verify Pod Health and Service Endpoints
kubectl get pods -n ml-training
# NAME                              READY   STATUS      RESTARTS   AGE
# model-serving-7f8d6c8b9-x4k2p     1/1     Running     0          45s
# model-serving-7f8d6c8b9-z9m1q     1/1     Running     0          45s
# pytorch-training-job-v9p2l        0/1     Completed   0          3m12s
```

```
kubectl port-forward svc/model-serving 8080:80 -n ml-training
curl -X POST http://localhost:8080/predict -F "image=@test_image.png"
```

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7. Project Reflection

Key Challenges & Learnings

Building this end-to-end pipeline was an insightful exercise in connecting practical deep learning with real-world infrastructure.

The most challenging part of the project was orchestrating the transition of model artifacts between training and serving inside Kubernetes. Training runs as a batch job that creates and updates model checkpoints, while serving runs continuously and needs immediate access to those saved weights. Setting up shared persistent volume claims (PVCs) and making sure the inference service didn't try to load incomplete weights during an active training epoch required careful coordination.

Another key learning point was configuring Kubernetes health probes and container permissions. Because PyTorch takes a few seconds to load the model into memory upon startup, early health checks can fail and cause pods to restart repeatedly. Adding an initial delay to the readiness probe solved this issue and ensured that traffic is only sent to pods once the model is fully loaded. Additionally, configuring non-root user permissions in Docker while retaining read access to mounted volumes highlighted important container security practices.

Overall, moving beyond local scripts to containerized workflows with automated CI and Kubernetes orchestration gave me a solid hands-on understanding of maintaining reliable, production-ready ML workloads.